The Song, Not the Pin: Route Transfer and Meaning-Based Place Identity in Collective Memory

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Abstract

Problem. The companion warp paper measures whether an agent can reach a target known only from a peer's memory. Two assumptions remain hidden in that construction — and in essentially all shared-memory multi-agent systems. First, what transfers is a *point*: broadcast snapshots carry place summaries but no edges, so the traveler learns WHERE and must re-derive HOW. Second, place identity is a *shared coordinate frame*: 'the same place' means 'close (x, y) in a grid the environment hands to every agent'; semantics only breaks ties inside a spatial radius. In the source culture a songline is neither: the song IS the route, and its places are identified by what they look like, not by coordinates nobody shares. **Approach.** We remove both assumptions and measure what each was hiding. (i) *Route warp*: broadcasts carry traversed edges with provenance; φ_{route} is the foreign share of edge mass along the committed path, and a strict $RW^{\star}$ is a connected foreign sub-path the traveler has never walked. (ii) *Semantic place identity*: places are matched by local semantic constellations (translation-invariant fingerprints); the inter-agent frame — translation in one step, full $SE(2)$ with rotations in the next — is recovered from mutually-unique matches of co-visited landmarks, and foreign evidence, including never-seen targets, is transported through it. **Findings.** What route knowledge buys is a step function, not a slope: place-only transfer collapses at the FIRST wall (success 1.0 at detour $D{=}1.00$ vs. 0.0 at $D{=}1.38$, and worse than blind exploration), while route transfer completes everywhere. Aging breaks the transferred route at a point predictable to the exact edge ($110/110$ regimes; all 40 mid-route ruptures across two world classes at the predicted index, error 0), and safety travels with the route: 0 hazard hits over the witness's path vs. 3.25 $[2.45, 4.10]$ for the place-only traveler, with the risk RESUMING at the predicted rupture point. Meaning-based identity recovers secret frames exactly (100% translation; $80/80$ under $SE(2)$), restores 88% of the shared-frame performance in the full peer stack, and fails *closed* under perceptual aliasing where the coordinate assumption fails *open* — and the companion distance law re-emerges tick-exactly (breakpoints $20/14/9$) with no shared origin and no shared north. **Thesis.** A shared map is not what makes collective navigation work. What transfers is the song — a sequence with meaning-anchored places — and each impoverishment of the song (point instead of route, coordinates instead of meaning) fails in a specific, measurable, and now predicted way. **Scope.** Grid worlds; translation and 90° rotations; measurement and minimal mechanisms, not a general theory.

1 Introduction

A songline encodes a journey: a sequence of places, each recognisable by what is there, bound into a path by the song's order. When multi-agent systems "share memory," they typically share something much poorer — a pin on a common map: coordinates of a target in a frame all agents are assumed to inhabit. The companion warp paper[1] made the act of navigating by someone else's memory a measured event ($W^\star$: a materialisation whose evidence is entirely foreign) and showed its value is governed by the coordination contract attached to the information. But its construction inherited the two impoverishments silently:

1. **The pin, not the song.** Broadcast snapshots carry place concepts — centroids, tags, confidences — but no edges. The receiver learns WHERE the water is; the route is re-derived from scratch. In open grids this is invisible: the manhattan path is the route. The moment paths are expensive, it is everything.
2. **Coordinates, not meaning.** 'Same place' is operationalised as coordinate proximity, with semantics as a tie-breaker inside a radius. The cross-agent place-correspondence problem — the reason map merging is hard in robotics — is bracketed by fiat.

This paper removes both assumptions, one at a time, with the measurement discipline of the series: every claim is registered before the runs, failures against registration are reported with their mechanisms, and the strongest results are exact-match validations of closed-form predictions.

Contributions.

1. **Route warp** ($RW^\star$, φ_{route}): edge-level provenance and the strict route-warp event — a connected foreign sub-path the traveler never walked (§3). The value of route over place transfer is a *cliff*, not a slope: place-only transfer dies at the first wall ($1.0 \to 0.0$ success between measured detour 1.00 and 1.38) and is WORSE than no transfer at all there, while route transfer completes everywhere (§3.1).
2. **The rupture law**: a witness stamps edges sequentially, so its trace ages non-uniformly and a delayed traveler chases a fading song. Replaying the series' trust×staleness gate over calibrated kinematics predicts *where the song breaks* to the exact edge: $110/110$ regime matches, all 21 mid-route ruptures at the predicted index (error 0), and a discrete trust-flip that zeroes route commits entirely (§4).
3. **Hazard stratification**: the wall and hazard axes are independent — walls break place-transfer's *completion*, hazards break its *safety* (0 vs. 3.25 hits at equal completion and $+0.7$ ticks), and when the song ruptures the risk resumes at the predicted edge (§5).
4. **Meaning-based place identity**: translation-invariant semantic fingerprints, mutually-unique landmark matching, and consensus frame recovery — translation first, full $SE(2)$ with secret $90°$ rotations after (§6). Exact frame recovery (100%; $80/80$ under $SE(2)$); online alignment in the full peer stack recovering 88% of the shared-frame gap; a proved failure asymmetry (semantic identity fails *closed* under aliasing and under in-principle-unrecoverable $180°$ symmetry; coordinate identity fails *open*); and the companion distance law reproduced tick-exactly $(20/14/9)$ with no shared frame at all (§6.1).

2 Setting

Grid worlds (12×10 to 14×12) with semantic water targets, hazards, and walls; agents with symbolic place memory, peer broadcast on cadence K, and the series' standard planner (target lock = the Q/R/M/C $M^\star$ event; completion $C^\star$). The trust×staleness inclusion rule of the collective-memory companion is used throughout with unchanged constants: evidence mass trust $\cdot\, e^{-\alpha\,\mathrm{age}} \cdot m$, threshold τ=0.30, α=0.05. Warp events follow the companion warp paper: provenance φ frozen at lock time; completion attributed to a continuous lock from a foreign commit (φ=1 at first lock) to success.

[1] Companion submissions: the Q/R/M/C measurement framework, the collective-semantic-memory architecture study, and the semantic-warp measurement paper. Citations deferred per concurrent-submission policy.

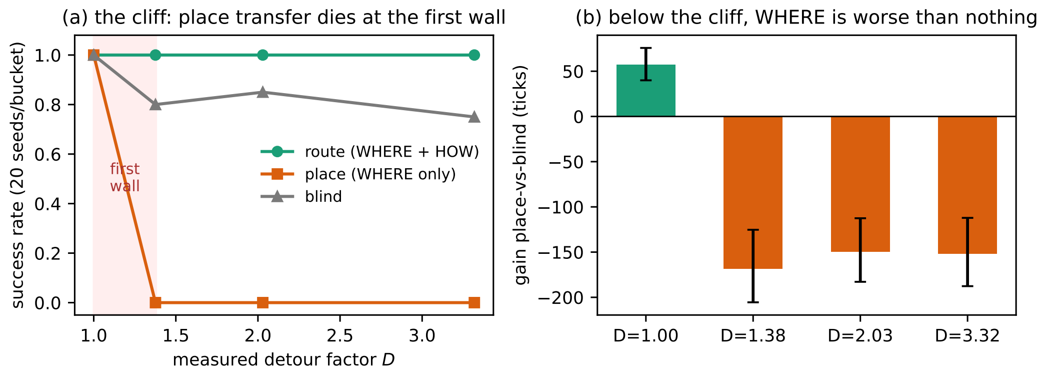

Figure 1: The cliff (paired arms on identical mazes, 20 seeds per measured-detour bucket). **(a)** Place-only transfer does not degrade with detour — it collapses at the FIRST wall, located by the probe bucket between D=1.00 and D=1.38; route transfer completes everywhere. **(b)** Below the cliff, knowing WHERE without HOW is *worse than knowing nothing*: the known-target manhattan pull pins the planner against walls (-150 to -168 ticks vs. blind exploration), while in the open world the same knowledge is worth $+57$.

3 Route warp: transferring the HOW

Edge provenance. Broadcast snapshots additionally carry the sender's OWN traversed edges $(a, b, \text{count}, \text{last tick})$; foreign edges are never re-broadcast (the same privacy contract as place snapshots: no agent reads another's graph, transport only). For the path P a planner commits,

$$\varphi_{\text{route}}(P) = \frac{\sum_{e \in P,\ \pi(e) \neq \text{self}} w(e)}{\sum_{e \in P} w(e)}, \qquad w(e) = \text{trust} \cdot e^{-\alpha\,\text{age}(e)} \cdot \log(1{+}\text{count}(e)),$$

the same weights the route gate uses. A **strict** $RW^\star$ is a committed, traversed, connected sub-path of ≥ 5 edges with $\varphi_{\text{route}} = 1$: the whole verse is someone else's. Place warp is the degenerate case (path length 0: only the target is foreign), giving a two-dimensional provenance plane $(\varphi_{\text{place}}, \varphi_{\text{route}})$.

First strict $RW^\star$. On a comb maze with measured detour D=2.65, a traveler commits and walks a fully foreign 45-edge route to completion in 54 ticks; with the same foreign PLACE evidence but no edges, the standard planner never arrives (300-tick budget), and blind exploration takes 126.

3.1 The cliff: what route knowledge buys

Table 1: Three arms on identical mazes (paired by seed), bucketed by MEASURED detour factor D; 20 seeds per bucket, censored gains (fail = 300). Place-only transfer does not degrade with D — it collapses at the first wall, and is worse than blind exploration there (the known-target manhattan pull pins the planner against walls). Route transfer completes everywhere; every completion is a strict $RW^\star$ with $\varphi_{\text{route}} = 1$.

D	gain route-vs-place [CI]	gain route-vs-blind [CI]	succ r/p/b
1.00	$+4.7$ $[+3.4, +6.0]$	$+62$ $[+44, +80]$	20/**20**/20
1.38	$+281$ $[+279, +283]$	$+113$ $[+77, +155]$	20/**0**/16
2.03	$+273$ $[+271, +275]$	$+123$ $[+91, +160]$	20/**0**/17
3.32	$+262$ $[+259, +264]$	$+110$ $[+72, +152]$	20/**0**/15

Figure 1 shows both faces of the result. Two pre-registered predictions failed here, in instructive ways. The negative control at D=1 was registered in ticks ($|\text{gain}| \leq 3$) and came out $+4.7$: the gap is *turn economy* (BFS routes walk straight legs; the dominant-axis planner alternates axes), not route knowledge — in path LENGTH the gap is exactly 0.0 $[0.0, 0.0]$ across all 20 open-grid cells. And "gain grows with D" is unmeasurable: with place success at 0 the gain saturates at the censoring ceiling. The correct form of the law is a step located by the probe bucket at $D \in (1.00, 1.38)$, i.e. the first wall.

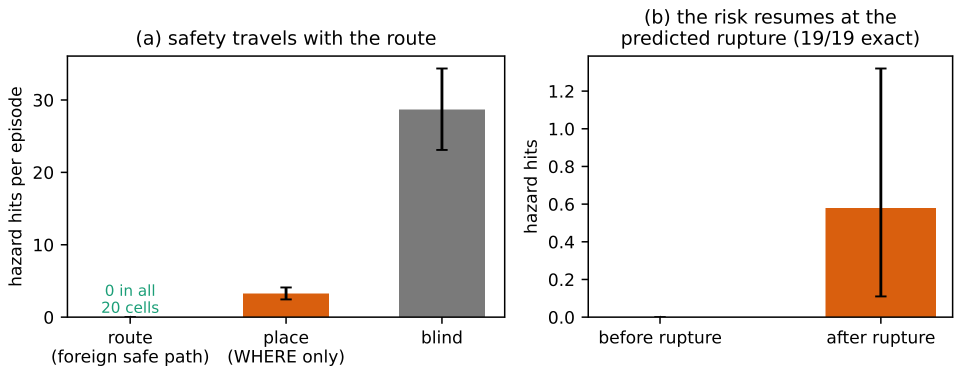

Figure 2: The song protects. **(a)** Hazard hits per episode: the traveler on the witness's route inherits its safety outright (0 in all 20 worlds); the place-only traveler pays 3.25 hits for the same completion; blind exploration pays 28.7. **(b)** Composition with the rupture law: in the rupture regime the traveler takes zero hits while the song lasts, and the risk resumes after the break — which occurs at the R2-predicted edge index in all $19/19$ cells.

4 The rupture law: where the song breaks

The witness stamps edge i at tick i: its trace ages non-uniformly, the start of the route always older than the end, and a traveler delayed by a_0 *chases a fading song*. Under the standard gate the binding edge is the CURRENT one (the oldest remaining), giving three regimes on the a_0 axis: complete; rupture at a predictable path index; dead on arrival. The predictor is fully deterministic — a no-decay calibration run records the traveler's kinematics (path index per tick, turns included), and the registered prediction replays the closed-form gate over that trajectory, the same device that produced the companion paper's $6/6$ hold-out.

Across 5 mazes $\times$ 11 delays $\times$ 2 trusts (110 cells, registered before execution): **110/110 regime matches; all 21 mid-route ruptures at exactly the predicted edge index (error** 0**)**; and a route trust-flip — at trust 0.7 the single-traversal edge mass gives $\mathrm{age}_{\max} = 9.6 < L$, so *no route ever commits* (0 commits observed; the discrete off-switch of the companion law, now for paths). The rupture of a transferred route is thus not noise but a law: the same constants $(\alpha, \tau, \mathrm{trust})$ that set WHEN a place-warp gate closes set WHERE a route-warp breaks.

5 Hazard stratification: the song protects

Table 2: Open fields dense with passable-but-punished hazards (density 0.20), the witness's route hazard-avoiding by experience; 20 paired worlds, decay off (the safety axis isolated from the aging axis). Both transfer modes COMPLETE — the wall axis (Table 1) and the hazard axis are independent; they differ in the price.

Arm	hazard hits [CI]	completion	mean t
route (foreign safe path)	**0** (all 20 cells)	20/20	16.6
place (WHERE only)	3.25 $[2.45, 4.10]$	20/20	15.9
blind	28.7	—	—

The traveler following the witness's path inherits its safety outright, at a time premium of $+0.7$ ticks (Figure 2a). Composing with §4: in the rupture regime (19/20 worlds) the traveler takes **0 hits before the rupture** and 0.58 $[0.11, 1.32]$ after it, when it falls back on place knowledge and re-enters the field — with all 19 ruptures again at the R2-predicted index (error 0; the edge-exact predictor replicated on a new world class with no adjustment). The song protects while it lasts, and it stops protecting exactly where the law says the verse ends.

6 Meaning-based place identity

Everything so far — and everything in the companion papers — rides on a shared coordinate frame. We now remove it: each agent's private frame is the true grid under a secret translation (later: plus a secret $90°$-multiple rotation), and broadcasts carry private coordinates that are meaningless to the receiver until it decides what the sender's places MEAN.

Fingerprints and the handshake. A place fingerprint is the local semantic constellation around a visited cell — which content tags (hazards, walls, water) sit at which relative offsets — translation-invariant and coordinate-free. Matching is *mutually unique* (a pair is accepted only if each side is the other's sole candidate above an exact-equality threshold; the sender knows far more places than the receiver, and one-directional uniqueness aliases distant look-alikes into the receiver's neighbourhood), fingerprints need ≥ 2 salient keys, and the frame translation is the modal delta of ≥ 3 agreeing pairs. Landmarks must carry at least one CONTENT tag: border-only ('void') patterns describe the shape of the world, not a place, and are excluded — the lesson of the corner trap below.

Witness–traveler results. Over 4 offsets $\times$ 2 hazard densities $\times$ 10 seeds: the semantic aligner recovers the secret offset *exactly in* 100% of landmark-world runs (mean 15.1 mutual pairs) and completes with strict $W^{\star}$ (φ=1) every time; the coordinate baseline under any nonzero offset locks a non-water cell in 100% of runs (its rare 'successes', 6.7%, are the displaced route accidentally crossing the true water). In the aliased world (featureless interior band) the two identity regimes break in OPPOSITE directions: semantic identity makes *zero* locks — it fails closed — while coordinate identity keeps confidently locking wrong cells — it fails open. This asymmetry, not raw accuracy, is the deepest argument for meaning-based identity: when the world does not support identification, the semantic system knows it.

6.1 The full stack, and removing the last axis

Online alignment in the peer stack. With N=4 agents under scarcity, all in distinct secret frames, alignment develops DURING the episode: agents explore, co-visited landmarks accumulate, and per-sender frames lock in mid-run (onset in 100% of episodes, mean tick $\approx$15). Pooled success: shared-frame oracle 0.456, coordinate under misalignment 0.350, semantic 0.444 — **88% of the gap recovered**, with strict $W^{\star}$ functioning through recovered frames.

The law without the frame. Re-running the companion distance law with the trust$\times$staleness gate ON TOP of semantic identity under a frame offset gives breakpoints $20/14/9$ — *the same integers as the shared-frame closed form, exactly*. The gate and the frame recovery are orthogonal layers: the law does not notice that the shared grid is gone.

$SE(2)$**: no shared north.** With secret rotations added, one rotation hypothesis per $90°$ reuses the translation aligner and the winner must *dominate* ($\geq 2\times$ the runner-up's consensus). Results: exact (r^{*}, δ^{*}) recovery in **80/80** rotation$\times$offset$\times$seed combinations with 100% strict-$W^{\star}$ completion; in a $180°$-symmetric world — where the frame is unrecoverable *in principle* — the aligner refuses ($0/10$ locks) while coordinates fail open ($10/10$ wrong locks); and the distance law again does not notice (bp $= 20$ under a $90°$-rotated witness). The translation-only aligner, for contrast, never recovers a rotated frame ($0/60$) but does not always fail closed: in $2/60$ cells a $180°$-symmetric hazard substructure lets it fail OPEN — the empirical motivation for the dominance rule.

The corner trap. In the full stack, raw success is non-discriminating under rotations (rotated coordinate poison mostly maps off-grid and degrades into pseudo-exploration), so the registered discriminator is *warp-lock precision*: the share of strict $W^{\star}$ locks pointing at real water. Before the void filter, early border fingerprints (agents start in corners; rectangle borders are $180°$-symmetric) fed wrong-frame hypotheses and semantic precision was 0.159; with borders excluded as non-landmarks it is 1.000 **(**$8/8$**) vs.** 0.000 **(**$0/992$ **phantom locks) for coordinates**, with alignment onset in 100% of episodes and clean regressions on all earlier experiments (Figure 3).

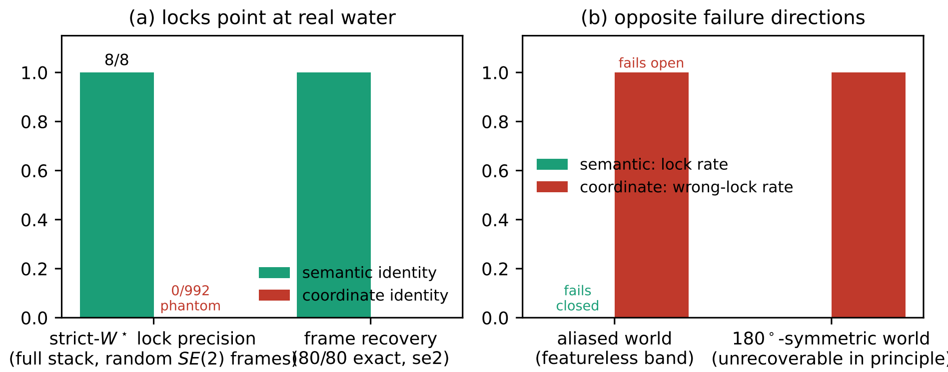

Figure 3: Meaning-based identity under $SE(2)$ frames. **(a)** Every strict-$W^\star$ lock the semantic stack makes points at real water $(8/8)$; the coordinate stack makes 992 locks, all phantoms $(0/992)$; frame recovery itself is exact in $80/80$ witness–traveler combinations. **(b)** The failure asymmetry in worlds where identification is impossible (featureless band; a $180°$-symmetric layout where the frame is unrecoverable in principle): semantic identity refuses to lock — fails closed — while coordinate identity keeps confidently locking wrong cells — fails open.

7 Related work

Map merging and place recognition. Multi-robot SLAM treats inter-map data association as an estimation problem over metric features Cadena et al. (2016); visual place recognition Lowry et al. (2016) matches appearance. Our contribution is not a competing matcher but the measurement layer on top: provenance-conditioned completion through recovered frames, the fail-closed/fail-open asymmetry as a registered prediction, and law invariance under frame removal.

Route sharing. Stigmergy transfers paths anonymously (pheromone gradients); learned communication (e.g. CommNet Sukhbaatar et al. (2016)) transfers policies opaquely. Neither exposes an edge-provenance decomposition, so neither can state — let alone predict — where a transferred route breaks. The rupture predictor is Time Warp Jefferson (1985) thinking applied to memory decay: deterministic replay of a gate over calibrated kinematics.

Songlines. The narrative frame is now load-bearing in both directions: the song is a sequence (route warp), its places are recognised by meaning (semantic identity), it fades verse by verse (rupture law), and it protects the walker while it lasts (hazard stratification).

8 Limitations and honest negatives

(1) Grid worlds; translation and $90°$ rotations only — continuous $SE(2)$ and appearance noise are future work (the fingerprint threshold is exact-equality because the substrate is noise-free; noisy substrates need the margin machinery of place recognition). (2) Four registered predictions failed and are reported with mechanisms: the $D{=}1$ negative control in ticks (turn economy; exact in path length), gain monotonicity in D (censoring; the truth is a cliff), the full-stack success comparison under rotations (non-discriminating; replaced by lock precision), and "translation-fails-closed" $(2/60$ fail open via symmetric substructures). (3) The route follower is a new planner tier; all three arms share it, but route-warp value is measured relative to this consumer. (4) Hazard safety transfers by construction when the witness's route is safe; the measured content is that place transfer CANNOT carry it and that the protection ends at the predicted rupture. (5) Scarcity-level completion of warp locks remains governed by the collision phenomenology of the companion paper.

9 Conclusion

Strip the song down to a pin on a shared map and two things break invisibly: the HOW (fatal at the first wall, expensive in every hazard field) and the WHO-MEANS-WHAT (fatal under any frame disagreement, and dangerous precisely because it fails open). Restore the song — edges with provenance, places with meaning — and both failures become measurable, lawful, and largely repaired: routes transfer completion and safety and break only where the aging law says; meaning recovers frames exactly, refuses honestly when the world is ambiguous, and carries the entire measurement stack — φ, gates, distance law — unchanged into a world where agents share no coordinates at all. The pin was never the memory. The song is.

A Registered predictions and revisions

All predictions were written to disk before execution; the artifact retains both versions wherever an operationalisation was revised. Revisions: R1 negative control re-measured in path length; R1 gain monotonicity re-formulated as cliff detection; W9 full-stack discriminator changed from success to warp-lock precision; W8 strict-$W^{\star}$ functioning extended to include warp-assisted own completions (taxi chains).

B Reproducibility

```
PYTHONPATH=. python experiments/warp/exp_route_warp_r0.py
PYTHONPATH=. python experiments/warp/exp_route_warp_r1.py --seeds 20
PYTHONPATH=. python experiments/warp/exp_route_warp_r2.py --seeds 5
PYTHONPATH=. python experiments/warp/exp_route_warp_r3.py --seeds 20
PYTHONPATH=. python experiments/warp/exp_warp_semantic_identity.py
PYTHONPATH=. python experiments/warp/exp_warp_semantic_stack.py --seeds 20
PYTHONPATH=. python experiments/warp/exp_warp_rotation.py --seeds 10
```

References

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